



BIOLOGICAL SYSTEMS MODELING AND ANALYSIS

Homework Assignment 5

【Spring Semester, 2025】

Due Date: May 28, 2026

Problem 1: (25%)

Calculate the expected variances for the following functions. For functions of two variables (x, y) , calculate both correlated and uncorrelated forms.

(1). $z = e^{k_1 x}$

(2). $z = k_1 \cos(k_2 x) + k_3 \sin(k_4 y)$

(3). $z = x^3 y^{-3}$

Problem 2: (25%)

Perform an individual parameter perturbation sensitivity analysis on the extinction model (Eq. 9.4). For each parameter, repeat the analysis using three perturbation levels: 2%, 10%, and 20% of the mean. Rank the parameters by sensitivity. Compare your conclusions to the Monte Carlo error analysis.

Problem 3: (25%)

Use local stability analysis to show that Case III (Fig. 9.11) of the Gause competition model is stable. Use the parameters given in the text (page 208), except let $K_2 = 800$ and $\beta = 3$.

Problem 4: (25%)

Consider the following system:

$$\frac{dx}{dt} = a_1 x^2 - a_2 x^3 - bxy$$

$$\frac{dy}{dt} = dxy - fy^2$$

with: $a_1 = 1$ $a_2 = 0.05$ $b = 5$ $d = 1$ $f = 10$

- (1). Write equations for and graph the nullclines of the above equations (i.e., do not non-dimensionalize) for $x \geq 0$ and $y \geq 0$. Show all the vectors associated with all the nullclines.



Assume all the parameters are positive. Show the algebra that proves the directions of the vectors.

- (2). On the graph, identify all the equilibria. Without considering the particular numerical values, write the coordinates of the equilibria in terms of the symbolic form of the parameters.
- (3). Using the parameters supplied above, perform a local stability analysis for all the equilibria.
- (4). Explain why your assessment of stability or instability is correct.

Notes:

1. Please submit your homework report together with your program to the NTU COOL course website.
2. Late submission will have a penalty of 10% discount per day of your homework total score toward a maximum of 50% discount. No late submission over five days will be accepted.